Performance Simulation Theory of Low-Level Wind Shear Detections Using an Airborne Coherent Doppler Lidar Based on RTCA DO-220

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***Abstract*— The performance simulation theory of low-level wind shear (LLWS) detection using an airborne coherent Doppler lidar (CDL) is shown. The simulation theory that this performance is based on is analogous to the turbulence and wind shear detections with an airborne Doppler radar, as specified in the radio technical commission for aeronautics (RTCA) document (DO)-220. Prior theoretical studies of CDL regarding: 1) signal-to-noise ratio (SNR) equation; 2) the relation between SNR and wind sensing performance; and 3) atmospheric parameters are fully utilized. An example simulation result under a practical condition satisfies the requirements to specify minimum operational performance standards (MOPS) in DO-220. Further, the simulation result is well aligned with past experimental results on the detectable range of LLWS. The results of this study can be utilized to establish MOPS for LLWS detections using an airborne CDL in the future.**

***Index Terms*— Airborne, aviation safety, DO-220, lidar, minimum operational performance standards (MOPS), turbulence, wind, windshear.**

I. Introduction

REMOTE wind sensing is an important factor for aviation safety at both the cruising and landing-approach altitudes. Considering its importance, the radio technical commission for aeronautics (RTCA) has published minimum operational performance standards (MOPS) for an airborne Doppler radar as document (DO)-220 [1] and its revised version [2]. This DO states the requirements for turbulence and wind shear detections with an airborne Doppler radar. This DO assumes only rainy or cloudy conditions, while windshear and turbulence under clear skies are not considered.

A coherent Doppler lidar (CDL) [3], [4] is attractive for the detection of invisible turbulence under clear sky conditions, including clear air turbulence (CAT), wake vortex, and low-level wind shear (LLWS). Therefore, many researchers have developed airborne CDLs and performed flight test campaigns. Bilbro et al. [5] reported the first airborne CDL using a wavelength of 10 μm in the 1980s. Targ et al. [6] presented the LLWS detections at go-around and landing-approach altitudes using 2-μm CDL in 1996. Hannon et al. [7] demonstrated the first CAT detection at high altitudes of up to 5 km in the late 1990s. Inokuchi et al. [8], [9], [10], [11], [12], [13] developed the first 1.5-μm airborne CDL in the 2000s and have continued to improve the system specification. These works have demonstrated many turbulence detections for CAT and LLWS.

Given these outcomes, RTCA performed a feasibility study toward standardization of airborne lidars, especially focusing on CAT detections at cruising altitudes. Unfortunately, the published feasibility study report [14] concluded that the existing CDLs were not able to meet the requirements of aircraft manufacturers for long-range CAT detections, mainly because of the very low-aerosol backscattering coefficient at cruising altitudes (around 10 km). However, it has also been stated that turbulence detections at lower altitudes are highly feasible. In fact, the demonstration of this application has been shown in the mid-1990s under the National Integrated Windshear program of the National Aeronautics and Space Administration (NASA) [6]. LLWS detections at about 7 km ahead of the aircraft have been demonstrated at an altitude of 500 m. In-flight experiments of this project, the aircraft encountered 13 LLWS cases with a windshear hazard index (referred to as F-factor) exceeding 0.06. The lidar-measured F-factor agreed to 0.01 for 12 cases and 0.02 for one case. Recently, the measurable range of 20 km at an altitude of 1.5 km or less has been demonstrated in the SafeAvio project of the Japan Aerospace Exploration Agency (JAXA) [13]. Forty-six experiments of a landing approach were performed to determine the performance of LLWS detections. In these experiments, an F-factor > 0.13 "must be detected," while an F-factor < 0.085 "must not be detected." Among the 13 LLWS events that should have been detected, the CDL showed 11 successful detections and four false alerts. The detectable ranges were 6.7–9.5 km at an altitude of 1000 m or less.

To establish MOPS for a CDL analogous to DO-220, sufficient evaluation in both simulation (see [1, Figs. 1–32]) and experimental demonstrations would be needed. Here, we show the performance simulation theory of LLWS detections using an airborne CDL. The simulation theory is analogous to that

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of Appendix B in RTCA DO-220 for Doppler radars [1]. The differences are in: 1) signal-to-noise ratio (SNR) equation; 2) relation between SNR and wind sensing performance; and 3) atmospheric parameters. The results of past CDL studies [14], [15], [16] are fully utilized in determining these terms and parameters. A simulation-based study on LLWS detections using an airborne CDL has also been published in [17], but this article is the first to study the relation with the currently used standard (i.e., DO-220). It should be noted that this article focuses on the performance study of a CDL.

While a direct detection lidar (DDL) [18], [19], [20], [21], [22] can utilize molecular backscattering, a CDL can only utilize aerosol backscattering. The molecular backscatter coefficient is higher than the aerosol backscatter coefficient at cruising altitudes. As a result, RTCA's feasibility study report [14] stated that DDLs have a theoretical advantage over CDLs for long-range CAT detections at cruising altitudes. Although some airborne DDLs have been reported in prior work [18], [19], [20], [21], [22] mainly for applications at cruising altitudes, DDLs are outside the scope of our study. LLWS detections are important at the go-around and landing-approach altitudes, which are part of the atmospheric boundary layer. In this atmospheric layer, the aerosol backscatter coefficient is much higher than the molecular backscatter coefficient. In such cases, the signal detection performance of the coherent (i.e., heterodyne) detection becomes higher than that of the direct detection. Therefore, it is obvious that CDLs are advantageous compared with DDLs for LLWS detections.

The rest of this article is organized as follows. The concept of LLWS detections using an airborne CDL is shown in Section II. The simulation theory for LLWS detection performance is described in Section III. In the same section, the analogy to DO-220 is summarized, along with the equations and input–output parameters of the simulation. In Section IV, an example of the performance simulation is introduced based on a practical CDL specification.

II. Concept of LLWS Detections Using an Airborne CDL

A. CDL Configuration

Fig. 1 shows the typical configuration of an airborne CDL. The detail of each component is not explained here since this is outside the scope of this article (for more detail, see [5], [6], [7], [8], [9], [10], [11], [12]). A continuous wave laser light is divided by a divider, and part of the divided light is sent to a balanced receiver through a coupler as a local light. The remaining light from the divider is pulsed by a modulator with a pulsewidth and a constant time interval. The pulsed light is amplified by an optical amplifier. The amplified light is transmitted to the atmosphere through a circulator, a telescope, and a scanner. The backscattered light from aerosols in the atmosphere is received through the scanner, the telescope, and the circulator. The received light is mixed with the local light by the coupler and heterodyne-detected by the balanced receiver. The heterodyne-detected signal is processed by a signal processor, and a range dependence of line-of-sight (LOS) wind velocity is estimated using range

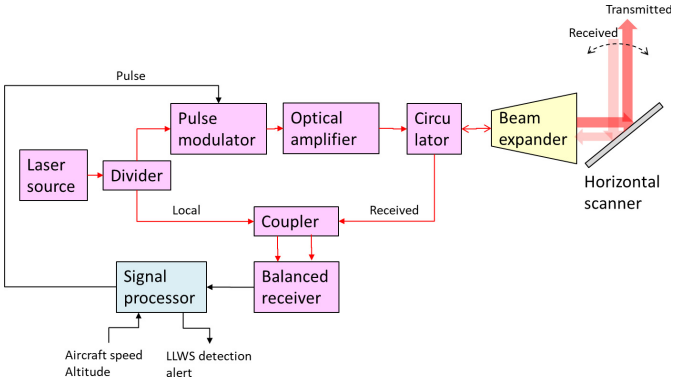


Fig. 1. Schematic of airborne CDL.

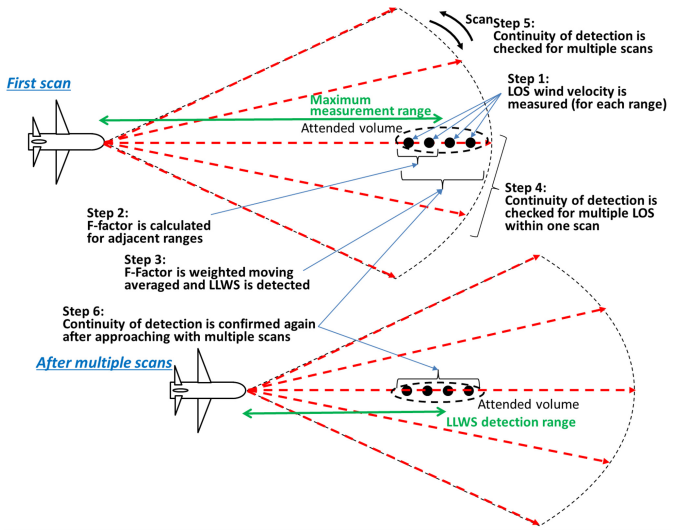


Fig. 2. Concept of LLWS detection.

gating and spectral analysis (i.e., the fast Fourier transform, spectral accumulation, spectral peak detection, and moment estimation). It should be noted that the CDL is airborne. Therefore, compensation of the aircraft speed regarding the LOS wind velocity is necessary for the signal processor. The above-mentioned operation of a CDL is performed for a horizontal scan, and an LOS wind distribution is obtained. Note that the influence of a lag angle, which is the angle difference between transmitting and receiving beams caused by an analog scan [23], is not considered here. This means the beam scan is performed step-by-step digitally. The signal processor performs not only LOS wind velocity estimates but also LLWS detections.

B. LLWS Detection

The concept of LLWS detection is based on RTCA DO-220 and is shown in Fig. 2. Since the exact definition of the wind field for LLWS is not defined by DO-220, we assume linear windshear [24] without loss of generality. A LOS wind velocity is estimated for each range and each direction (step 1 in the figure). The F-factor is calculated using the LOS velocities of two adjacent ranges (step 2).

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This is given by [25]

$$F = \frac{\Delta v_{\mathrm{LOS}}}{\Delta G}\left(\frac{V}{g} + \frac{2h_a}{V}\right) \tag{1}$$

where Δv_{LOS} is the difference of the LOS wind velocity between the adjacent two range gates (m/s), V is the aircraft speed (m/s), h_a is the aircraft altitude (m), g is the acceleration due to gravity (m/s^2), and ΔG is the range gate width (m). In the signal processing of heterodyne-detected signals, range gating with constant time interval without any overlaps and gaps is assumed. Therefore, ΔG is also the wind measurement range interval.

Then, a weighted moving average of the F-factor is computed along the LOS direction. An LLWS detection is performed using the averaged F-factor by comparing it with a predetermined threshold (step 3). The whole region of the moving averaged volume is referred to as an attended volume in the following. The above-mentioned process is done for each LOS. Continuity of the LLWS detections is checked for multiple LOSs within one scan, and it is compared with the threshold for this continuity (step 4). Further, the continuity is also checked for multiple scans regarding the attended volume, and it is compared with the threshold regarding the number of multiple scans (step 5). The above-mentioned steps (1–5) are iterated until the distance between the aircraft and the attended volume becomes the LLWS detection range after additional multiple scans (step 6). The continuity of detections is also checked in step 6. A detection alert is triggered if an LLWS is detected in all steps 3–6. Note that the attended volume is approaching depending on the aircraft speed and the detection performance should be evaluated at the LLWS detection range (i.e., the range from the airplane after the multiple scans). The possible scan number for the LLWS detection is determined by: 1) the maximum measurement range; 2) the aircraft speed; 3) the scanning speed; and 4) the scanning width. The set of these parameters infers a complicated trade-off in LLWS detection performance. For example, if the maximum measurement range is set at a long range, the distance between the attended volume and the evaluated range becomes long. In such a case, it takes a long time for the attended volume to reach the evaluated range. This results in a large number of possible scans. This could help improve LLWS detection performance, but it may also decrease performance since wind sensing performance at far ranges is generally worse. A high-scanning speed is also good for a large number of possible scans, but this degrades the angular resolution of the measurements and makes the number of LOS measurements in one scan smaller. These effects are considered in the simulation theory described in Section III.

III. Simulation Theory on LLWS Detection Performance

A. Simulation Procedure

The simulation procedure for LLWS detection is schematically shown in Fig. 3. The procedure is analogous to DO-220. A range dependence of SNR (here, wideband SNR) is calculated using CDL parameters and atmospheric parameters. The range-dependent wind sensing performance is calculated in the next step using SNR, signal processing parameters, and a wind field parameter (i.e., velocity width caused by different wind speeds within a range gate). Then an LLWS detection performance is simulated using the wind sensing performance, the CDL measurement parameters, and aircraft parameters. The detail of each step is shown in Sections III-B–III-D. Furthermore, the detail of the input and output parameters of the simulation are shown in Section III-E.

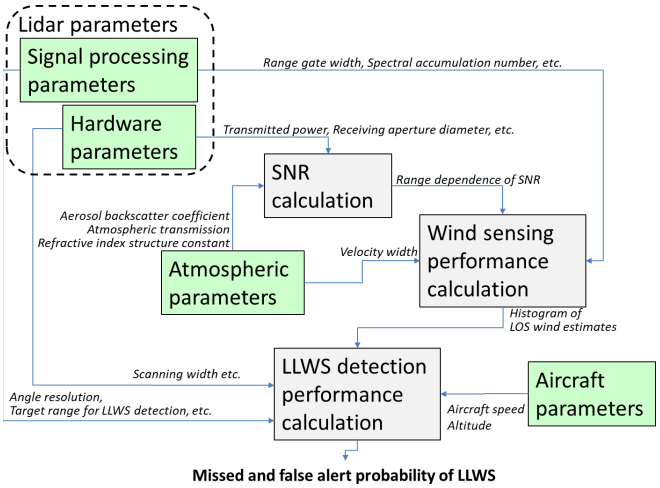


Fig. 3. Schematic of simulation procedure.

B. SNR Calculation

The range-dependent SNR of a CDL is calculated by [15]

$$\mathrm{SNR}(R) = \frac{\eta\eta_D(R)\lambda E\beta e^{-2(R_L\beta+\alpha_m)R}\pi D^2}{8hBR^2} \tag{2}$$

where h is Planck's constant (Js), R is the range (m), η is the system efficiency, λ is the wavelength (m), B is the wideband receiving bandwidth (Hz), E is the pulse energy (J), D is the receiving aperture diameter (m), β is the aerosol backscatter coefficient (/m/sr), R_L is the extinction to aerosol backscatter ratio (sr), and α_m is the molecular absorption coefficient (/m). $\eta_D(R)$ is the heterodyne-detection efficiency which is expressed by [15]

$$\eta_D(R) = \frac{\eta_F}{1 + \left(1 - \frac{R}{F_F}\right)^2\left[\frac{\pi(A_C D)^2}{4\lambda R}\right]^2 + \left(\frac{A_C D}{2S_0(R)}\right)^2} \tag{3}$$

where F_F is the beam focusing range (m), A_C is the correction factor of the receiving aperture, which is obtained by the nearest Gaussian approximation, and η_F is the far-field heterodyne-detection efficiency (i.e., the efficiency at the beam focusing range). In the case of the optimum beam truncation ratio (the ratio of the $1/e^2$ intensity beam diameter and the receiving aperture diameter) of 0.8, the values of A_C and η_F are 0.71 and 0.4, respectively [15]. $S_0(R)$ is the transverse coherence length of the received light (m). In the case of a horizontal path measurement with constant atmospheric parameters, this

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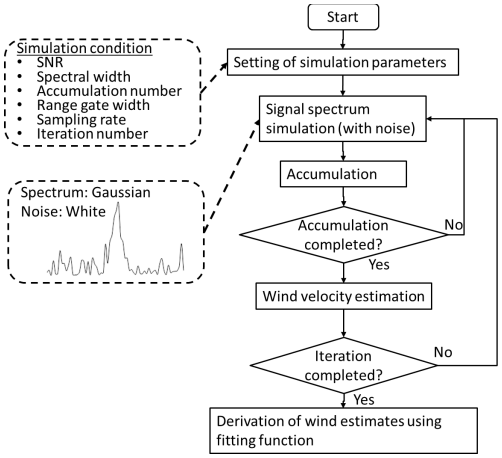


Fig. 4. Schematic of simulation for the relation between SNR, wind velocity, estimation precision, and signal detection probability.

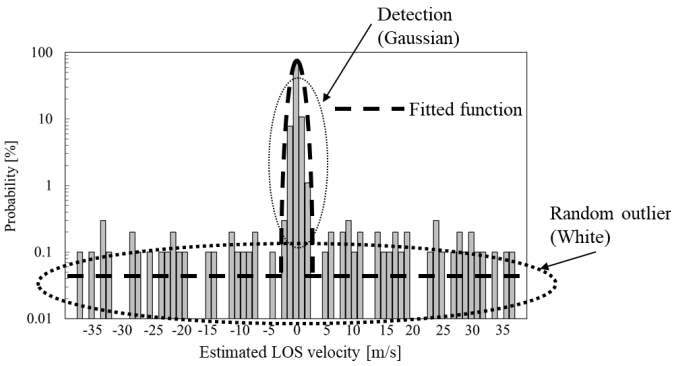


Fig. 5. Schematic of histogram of LOS wind velocity estimates at a certain SNR with fit function.

is expressed by [15]

$$S_0(R) \cong \left[1.1\left(\frac{2\pi}{\lambda}\right)^2 R C_n^2\right]^{-\frac{3}{5}} \tag{4}$$

where C_n^2 is the atmospheric refractive index structure constant ($m^{-2/3}$).

C. Wind Sensing Performance Calculation

In DO-220, the relation between SNR and wind velocity estimation precision has been expressed analytically for the case of a Doppler radar (see [1, eq. (2)]), but the above-mentioned relation has not been expressed analytically for the case of a CDL. Therefore, a Monte-Carlo simulation is employed here to show this relation. The simulation procedure is basically the same as in [16] and schematically shown in Fig. 4. This simulation method is selected because of its generality and simplicity, although there are more complicated methods for this kind of Monte-Carlo simulation (see [26], [27]). The noise is white Gaussian. The signal spectrum has Gaussian shape. The width (full width at half maximum (FWHM), Hz) is given by

$$B_S = \sqrt{2\ln(2)} \cdot \sqrt{\left(\frac{\sqrt{2\ln(2)}}{\pi\,\Delta t}\right)^2 + \left(\frac{4\Delta v}{\lambda}\right)^2 + \left(\frac{2\Delta f}{\sqrt{2\ln(2)}}\right)^2} \tag{5}$$

where Δt is the pulsewidth (FWHM, s), Δf is the laser linewidth (FWHM, Hz). Δv is the wind velocity width within the range gate (standard deviation, m/s) along with the LOS. This equation is equivalent to [17, eq. (19)] with additionally considering the laser linewidth. The wind velocity width depends on strength of the LLWS with the linearly chirping wind field [24]. In addition, the influence of the Kolmogorov turbulence [27] is considered in the wind velocity width. In the case of equal values between the range gate width and the range measurement interval, the wind velocity width is approximately given by utilizing (1) of this article and [27, eq. (2)], and expressed by

$$\Delta v \approx \frac{1}{\sqrt{12}} \cdot \frac{F}{\frac{V}{g} + \frac{2h_a}{V}} \cdot \Delta G + \sqrt{\frac{9}{20}} \cdot \varepsilon^{\frac{1}{3}} \cdot \Delta G^{\frac{1}{3}} \tag{6}$$

where ε is the energy dissipation rate (m^2/s^3). The first term of this equation corresponds to the LLWS and the second one corresponds to the Kolmogorov turbulence. This equation is expressed by a summation (not a root sum square) because the assumed LLWS is a linearly chirping wind field and deterministic. Further, it should be noted that a linearly chirping wind field has a uniform distribution along the LOS. Therefore, a component of the LLWS has a standard deviation of $y/(12)^{1/2}$ (y: difference of LOS wind velocity of an LLWS component between both sides of a range gate).

The spectral shape of the digital signal is determined by the LOS wind velocity width, pulsewidth, laser linewidth, sampling rate of the analog-to-digital (AD) conversion, and range gate width (corresponding to the range resolution). The noise is randomly generated by a computer simulation and added to the simulated signal spectrum with the level depending on the SNR. Wind velocity estimation is performed on this simulated spectrum with noise after incoherent spectral accumulations of a predetermined number. The accumulation number is given by

$$N = T \cdot \mathrm{PRF} \tag{7}$$

where T is the LOS measurement rate (s) and PRF is the pulse repetition frequency (PRF, Hz).

This simulation of LOS wind velocity estimation is performed iteratively. Statistical performance of the wind velocity estimation precision and signal detection probability is obtained for a certain SNR. The relation between SNR, wind velocity estimation precision, and signal detection probability can be derived by incrementally changing the SNR.

A histogram of wind velocity estimates at a certain SNR is shown in Fig. 5. It is known that there are two components, good estimates, and random outliers [26], [27]. The good estimates are basically Gaussian, while the outliers are white. Therefore, the histogram can be approximated with

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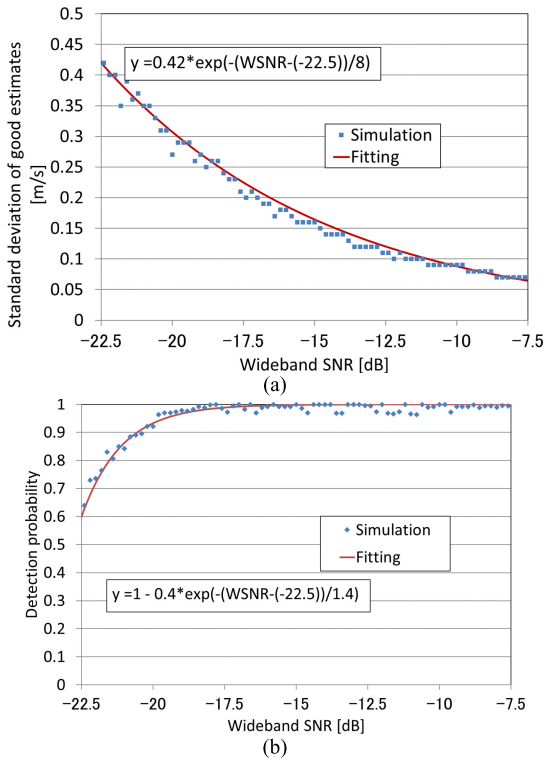


Fig. 6. Schematic of relation between SNR and (a) wind velocity estimation precision (standard deviation of good estimates) and (b) signal detection probability with fit function under the condition of laser wavelength: 1.55 μm, range gate width: 100 m, velocity width: 1 m/s, pulsewidth: 500 ns, laser linewidth: 250 kHz, sampling frequency: 100 MHz (complex sample), accumulation number: 100, and velocity search range from −30 to 30 m/s.

a least-square fitting using the following function [27]:

$$\mathrm{PDF}(v_e, \mathrm{SNR}(R)) = \frac{b(\mathrm{SNR}(R))}{v_{\mathrm{search}}} + \frac{(1-b(\mathrm{SNR}(R)))}{(2\pi)^{1/2} g_s(\mathrm{SNR}(R))} \exp\left[-\frac{(v_e - v_m)^2}{2g_s^2(\mathrm{SNR}(R))}\right] \quad (8)$$

where b is the probability of bad estimates and g_s is the standard deviation of good estimates (m/s), v_m is the true value of mean LOS wind velocity (m/s), v_e is the estimated one (m/s), and v_{search} is the velocity search range within the spectrum (m/s). The relation between SNR, wind velocity estimation precision, and signal detection probability (i.e., $=1-b$) with the fitting functions is shown in Fig. 6. An empirical fitting function is employed here to determine analytical formulas to express these relations. The fitting functions for the signal detection probability and the estimation precision are given as follows:

$$P_{\mathrm{rob}}(R) = 1 - a' \cdot \exp\left(-\frac{\mathrm{WSNR} - e'}{b'}\right) \quad (9)$$

$$P_{\mathrm{rec}}(R) = c' \cdot \exp\left(-\frac{\mathrm{WSNR} - e'}{d'}\right) \quad (10)$$

where WSNR is the wideband SNR (in the unit of dB). a'–d' are the fitting parameters. e' is the SNR (in the unit of dB) at which the detection probability is 0.6. Regarding the performance curve of (9) and (10), the SNR is equivalently improved approximately proportional to $\sqrt{N}$ if $N > 10$ [27].

D. LLWS Detection Performance Calculation

The standard deviation of the measured F-factor in a single LOS is given by [1]

$$\sigma_F(\mathrm{SNR}(R)) = \frac{3.464\sigma_v(\mathrm{SNR}(R))\left(\frac{V}{g} + \frac{2h_a}{V}\right)}{\Delta G\sqrt{(N_e+1)(N_e-1)N_e}} \quad (11)$$

where $\sigma_v(\mathrm{SNR}(R))$ is the standard deviation of the LOS wind velocity estimation (m/s), and N_e is the number of range gates used in the F-factor calculation. This equation corresponds to step 2 in Fig. 2.

In DO-220, the estimated F-factor is additionally processed using the weighted moving average within a certain range width, concretely, 1 km. This calculation is performed by [1]

$$\sigma_{FA}(\mathrm{SNR}(R)) = \sqrt{\frac{12}{N_a^2\sqrt{(N_e+1)(N_e-1)N_e}} \cdot \left[\sum_j \left(\sum_i C_i\right)^2_j\right]} \cdot \sigma_F(\mathrm{SNR}(R)) \quad (12)$$

where N_a is the number of calculated F-factors used for the moving average, C_i is the weighting of each range gate, i is the range bin number in the N_e ranges, and j is the number in the averaging of N_a times. This equation corresponds to step 3 in Fig. 2. It should be noted that a constant value of $\sigma_F(\mathrm{SNR}(R))$ is assumed within the moving averaging width (here, 1 km). The values for $\sigma_v(\mathrm{SNR}(R))$ and $\mathrm{SNR}(R)$ are approximately constant within the moving averaging width. The means to determine the weighting C_i is shown in Appendix A.

The calculation of F-factor performs the subtraction of two wind velocity estimations of the adjacent range rates [see (1)]. This means the histogram of the estimated F-factor has the following three components under the condition of $V_{\mathrm{search}} \gg g_s$. This condition is generally true for CDLs (see [27]).

1) Gaussian distribution corresponding to the case of two good estimates

$$\text{Probability: } [1-b(\mathrm{SNR}(R))]^2.$$

2) Triangle distribution corresponding to the case of two bad estimates

$$\text{Probability: } b(\mathrm{SNR}(R))^2.$$

3) Approximate triangle distribution corresponding to the case of one good estimate and one bad estimate

$$\text{Probability: } 2b(\mathrm{SNR}(R)) \cdot [1-b(\mathrm{SNR}(R))].$$

The standard deviation of the triangle distribution is $q/(18)^{1/2}$ (q: half width of the triangle distribution). The histogram of the measured F-factor can be expressed using 1)–3). This histogram is given by (13), as shown at bottom of the next page, where F_e and F_m are the estimated and true values of the F-factor. The value of F_m

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is set to zero in the simulation of this article for generalization. The symbols $\sigma_{\mathrm{FAGG}}(\mathrm{SNR}(R))$, $\sigma_{\mathrm{FAOO}}(\mathrm{SNR}(R))$, and $\sigma_{\mathrm{FAGO}}(\mathrm{SNR}(R))$ are the standard deviations of the estimated F-factor (weighted averaged) for the cases of 1)–3), respectively. The values of these symbols are calculated using (11) and (12). The value of $\sigma_v(\mathrm{SNR}(R))$ in (11) is calculated for each case as follows. For case 1), this value is g_s in (8). For case 2), it is $V_{\mathrm{search}}/(12)^{1/2}$ since the distribution of the bad estimates is white. For case 3), it is approximately considered as $V_{\mathrm{search}}/(2(24)^{1/2})$ by considering the sufficiently low deviation of the good estimates than that of the bad estimates (i.e., under the condition of $V_{\mathrm{search}} \gg g_s$).

With the additional assumption that the range dependence of SNR is frozen in one horizontal beam scan, the detection probability of the hazardous F-factor in a single LOS is calculated by [1]

$$PD^S(F_a, \mathrm{SNR}(R)) = \int_{F_T}^{\infty} \mathrm{PDF}_F(F, F_a, \mathrm{SNR}(R))dF \quad (14)$$

where F_a is the must-alert F-factor and F_T is the detection threshold of the hazardous F-factor. The probabilities of the missed and false alerts are calculated by [1]

$$PM^S(F_a, \mathrm{SNR}(R)) = 1 - PD^S(F_a, \mathrm{SNR}(R)) \quad (15)$$

$$PF^S(F_{na}, \mathrm{SNR}(R)) = \int_{FT}^{\infty} \mathrm{PDF}_F\left(F, F_{n_a}, \mathrm{SNR}(R)\right)dF \quad (16)$$

where F_{na} is the must-not alert F-factor. Equations (14)–(16) correspond to step 3 in Fig. 2.

In DO-220, the detection of the hazardous F-factor in a single LOS does not trigger an alert. Instead, the alert is generated when the detections appear continuously for the adjacent two or more LOSs. In this concept, the probabilities of the missed and false alerts for multiple LOSs are expressed by [1]

$$PM(F_a, \mathrm{SNR}(R)) = 1 - \left[PD^S(F_a, \mathrm{SNR}(R))\right]^M \quad (17)$$

$$PF(F_{na}, \mathrm{SNR}(R)) = \left[PF^S(F_{na}, \mathrm{SNR}(R))\right]^M \quad (18)$$

where M is the number of LOSs used in the judgment of the hazardous F-factor detection. In other words, these equations mean the probabilities that the missed and false alerts appear continuously for the adjacent M measurement directions. The value of M is determined by

$$M = \frac{\theta_r}{\theta_V \cdot T} \quad (19)$$

where θ_T is the angle resolution of the hazardous F-factor detection (°), θ_V is the beam scanning speed (°/s). Equation (19) is different from the corresponding equation in DO-220 by considering the difference in the beam divergence angle between a CDL and a Doppler radar. Equations (17)–(19) correspond to step 4 in Fig. 2.

For more than one scan (i.e., multiple scans), the alert and missed alert probabilities of the must-alert F-factor are calculated at a range R from the aircraft by [1]

$$PD'(F_a, R) = \prod_{k=1}^{N_S}[1 - PM(F_a, \mathrm{SNR}(R_k))] \quad (20)$$

$$PM'(F_a, R) = 1 - PD'(F_a, R) \quad (21)$$

where N_S is the number of scans. R_k is the range at the kth scan (m), and is derived by

$$R_k = R + \frac{\theta_W}{\theta_V} \cdot V \cdot (N_S - k) \quad (22)$$

where θ_w is the azimuth scan width (°). This equation expresses the approaching effect of the aircraft speed to the range R. In the case of reciprocating scans, $N_S = 2$ for one reciprocation. In this case, the CDL measures the adjacent range (far side) of the range R at the first scan and measures the range R at the second scan. The false alert probability of

$$\mathrm{PDF}_F(F_e, F_m, \mathrm{SNR}(R)) = \begin{cases} \dfrac{b(\mathrm{SNR}(R))^2 \cdot \left(\sqrt{18}\sigma_{\mathrm{FAOO}}(\mathrm{SNR}(R)) - |F_e - F_m|\right)}{18\sigma^2_{\mathrm{FAOO}}(\mathrm{SNR}(R))} \\ \quad + \dfrac{2b(\mathrm{SNR}(R)) \cdot \left(\sqrt{18}\sigma_{\mathrm{FAGO}}(\mathrm{SNR}(R)) - |F_e - F_m|\right)}{18\sigma^2_{\mathrm{FAGO}}(\mathrm{SNR}(R))} \\ \quad + \dfrac{[1 - b(\mathrm{SNR}(R))]^2}{(2\pi)^{\frac{1}{2}}\sigma_{\mathrm{FAGG}}(\mathrm{SNR}(R))} \exp\left[-\dfrac{(F_e - F_m)^2}{2\sigma^2_{\mathrm{FAGG}}(\mathrm{SNR}(R))}\right], & \text{if } |F_e - F_m| < \sqrt{18}\sigma_{\mathrm{FAGO}}(\mathrm{SNR}(R)) \\ \dfrac{b(\mathrm{SNR}(R))^2 \cdot \left(\sqrt{18}\sigma_{\mathrm{FAOO}}(\mathrm{SNR}(R)) - |F_e - F_m|\right)}{18\sigma^2_{\mathrm{FAOO}}(\mathrm{SNR}(R))} \\ \quad + \dfrac{[1 - b(\mathrm{SNR}(R))]^2}{(2\pi)^{\frac{1}{2}}\sigma_{\mathrm{FAGG}}(\mathrm{SNR}(R))} \exp\left[-\dfrac{(F_e - F_m)^2}{2\sigma^2_{\mathrm{FAGG}}(\mathrm{SNR}(R))}\right], & \text{if } \sqrt{18}\sigma_{\mathrm{FAGO}}(\mathrm{SNR}(R)) \\ & \quad \le |F_e - F_m| < \sqrt{18}\sigma_{\mathrm{FAOO}}(\mathrm{SNR}(R)) \\ \dfrac{[1 - b(\mathrm{SNR}(R))]^2}{(2\pi)^{\frac{1}{2}}\sigma_{\mathrm{FAGG}}(\mathrm{SNR}(R))} \exp\left[-\dfrac{(F_e - F_m)^2}{2\sigma^2_{\mathrm{FAGG}}(\mathrm{SNR}(R))}\right], & \text{if } |F_e - F_m| \ge \sqrt{18}\sigma_{\mathrm{FAOO}}(\mathrm{SNR}(R)) \end{cases} \quad (13)$$

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the must-not alert F-factor regarding continuous detections for N_s scans is expressed by [1]

$$PF'(F_a, R) = \prod_{k=1}^{N_S} PF(F_{na}, \mathrm{SNR}(R_k)). \tag{23}$$

It should be noted that the above-mentioned equations show the process regarding the detection of a hazardous F-factor, but the LLWS detection alert is not triggered at the above-mentioned steps in DO-220. Equations (20)–(23) correspond to step 5 in Fig. 2.

The LLWS detection alert is triggered when hazardous alerts are detected continuously for multiple sets of N_S scans. The range where the hazardous F-factor has been detected depends on the aircraft speed. The number of possible scans for this range, until it reaches an LLWS detection range, is calculated by

$$N(R) = \left\lfloor \frac{R - R_S}{V \cdot \frac{N_S \cdot \theta_W}{\theta_V}} \right\rfloor + 1 \tag{24}$$

where R_S is the LLWS detection range (m), which is a required value in the MOPS. Note that the signal detection probability at the range R must be higher than the predetermined threshold of wideband SNR. This threshold SNR can be derived analytically using (9). The missed alert probability (i.e., probability for no detection at all) in $N(R)$ scans for the must-alert F-factor is expressed as [1]

$$PM''(F_a, R) = \prod_{k=1}^{N(R)} PM'(F_a, R_k). \tag{25}$$

In a similar manner, the false alert probability for the must-not alert F-factor is expressed as [1]

$$PF''(F_{na}, R) = 1 - \prod_{k=1}^{N(R)} \left[1 - PF'(F_{na}, R_k)\right]. \tag{26}$$

In DO-220, the appearance probability of the F-factor regarding the population of LLWS (i.e., microburst) is given by [1]

$$f_H(F) = \frac{2F \cdot \exp\left[-\frac{F}{0.123^2}\right]}{0.123^2}. \tag{27}$$

The missed alert probability of hazardous LLWS detection averaged over the population is calculated by a convolution of (25) and (27). This is given by

$$\langle PM(R) \rangle = 1 - \frac{\int_{\mathrm{FDLO}}^{\mathrm{FDHI}} f_H\left(\hat{F}_a\right)\left[1 - PM''\left(\hat{F}_a, R\right)\right] d\hat{F}_a}{\int_{\mathrm{FDLO}}^{\mathrm{FDHI}} f_H\left(\hat{F}_a\right) d\hat{F}_a} \tag{28}$$

where FDLO and FDHI are the minimum and maximum of the hazardous F-factors in the LLWS population. This equation is a little different from the corresponding equation of DO-220 [1, eq. (22)] in which the appearance probability function of the atmospheric reflectivity is included. The performance of a CDL is influenced not only by the atmospheric reflectivity (aerosol backscatter coefficient in the case of a CDL) but also by several parameters, for example, refractive index structure constant, etc. This is the reason for the above-mentioned difference. The appearance probability of the atmospheric parameters is not considered in (28). Instead, the atmospheric parameters with a high-cumulative appearance probability of 90% are used in the simulation.

In the same manner, the false alert probability of non-hazardous LLWS is given by

$$\langle PF(R) \rangle = \frac{\int_{\mathrm{FFLO}}^{\mathrm{FFHI}} f_H\left(\hat{F}_a\right) PF''\left(\hat{F}_a, R\right) d\hat{F}_a}{\int_{\mathrm{FFLO}}^{\mathrm{FFHI}} f_H\left(\hat{F}_a\right) d\hat{F}_a} \tag{29}$$

where FFLO and FFHI are the minimum and maximum of the non-hazardous F-factors in the LLWS population. Equations (24)–(29) correspond to step 6 in Fig. 2.

E. Input and Output Parameters of Simulation

The input and output parameters of the simulation are summarized in Tables I and III. The aircraft altitude of 1600 ft in Table I is set as the maximum altitude of LLWS which has been defined in [28]. The molecular absorption coefficient has been derived using the U.S. standard atmosphere model. Other atmospheric parameters are the ones denoted in the RTCA's feasibility study of airborne lidars [14]. The values correspond to the cumulative appearance probability of 90%. The wind velocity width of 1 m/s in Table I corresponds to the maximum hazardous F-factor and the aircraft parameters in the table. The energy dissipation rate of 0.006 $\mathrm{m}^2/\mathrm{s}^3$ is additionally assumed to set the wind velocity width using (6). This value is much lower than the ones in [27] (0.014, 0.032, and 0.322 $\mathrm{m}^2/\mathrm{s}^3$) which correspond to low, moderate, and severe Kolmogorov turbulences. However, the scope of this article is the detection of LLWS under usual Kolmogorov turbulences. Therefore, the above-mentioned energy dissipation rate is a reasonable value in the atmospheric boundary layer (for example, [29, Fig. 5]). Ideally, the wind velocity width should be calculated for each F-factor in the integral computation of (28) and (29). However, this ideal concept makes the simulation too complicated. This is because the time-consuming Monte-Carlo simulations of Fig. 4 become necessary for all wind velocity widths. Therefore, the simulation is performed here with a fixed wind velocity width. This concept of simulations with a fixed wind velocity width is also analogous to DO-220. The hardware parameters in Table I are based on a CDL that has been developed in the SafeAvio project [11], [12], [13] with some empirically reasonable modifications. The system efficiency in Table I has been set by considering decrease of the efficiency in the installation process of a CDL on an airplane. It should be noted that an empirical value of system efficiency of a ground-based CDL [23] has been denoted as the same value as in Table I (i.e., 0.1), but the definition of the efficiency in [23] includes the far-field heterodyne-detection efficiency of 0.4. On the other hand, the far-field heterodyne-detection efficiency is considered separately from the system efficiency in this article [see (2)]. Therefore, the system efficiency of this article is 0.4 times lower than the value in [23] under the same definition. The signal processing parameters are based on DO-220, especially parameters related to the hazardous F-factor and LLWS detection. The correspondence between the equation in this article and that of DO-220 is also shown

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TABLE I
INPUT PARAMETERS OF SIMULATION

Parameters			Symbol	Value	Unit
Atmosphere		Aerosol backscatter coefficient	β	1.1×10^{-7}	/m/sr
		Extinction to aerosol backscatter ratio	R_L	50	sr
		Molecular absorption coefficient	α_m	5.4×10^{-6}	/m
		Atmospheric refractive index structure constant	C_n^2	8.9×10^{-16}	$m^{-2/3}$
		Wind velocity width	Δv	1	m/s
		Appearance histogram of F-factor	$f_H(F)$	Eq. (27) *2	
Aircraft		Aircraft speed	V	80 *2	m/s
		Altitude	h_a	488 (1,600 ft)	m
CDL	Hardware	Laser wavelength	λ	1.55×10^{-6}	m
		Laser linewidth	Δf	2.5×10^{3}	Hz
		Pulse energy	E	3×10^{-3}	J
		Pulse width	Δt	5×10^{-7}	s
		Pulse repetition frequency	PRF	1,000	Hz
		Receiving aperture diameter	D	0.11	m
		Aperture correction factor	A_C	0.71	
		Beam focusing range	F_F	20,000	m
		Receiving bandwidth (wideband)	B	1×10^{8}	Hz
		Sampling rate (in complex sample)	-	1×10^{8}	Hz
		System efficiency	η	0.1	
		Far-field heterodyne-detection efficiency	η_F	0.4	
		Beam scanning speed	θ_V	5 *1	deg/s
		Scanning width	θ_w	50 *1 (+/- 25)	deg
	Signal processing	Must alert F-Factor	F_a	0.13 *1	
		Must-not alert F-Factor	F_{na}	0.085 *1	
		Hazardous F-factor alert threshold	F_T	0.105 *1	
		Minimum of the hazardous F-factor	$FDLO$	0.13 *2	
		Maximum of the hazardous F-factor	$FDHI$	0.3 *2	
		Minimum of the non-hazardous F-factor	$FFLO$	0 *2	
		Maximum of the non-hazardous F-factor	$FFHI$	0.085 *2	
		Number of range gates used in the F-factor calculation	Ne	5 *1	
		Number of range gates for the calculation of weighted moving averaged F-Factor	N_a	7 *1	
		Angle resolution of hazardous F-factor detection	θ_T	5	deg
		Number of scans in the judgement of multiple scan	N_S	2	
		Range gate width	ΔG	100	m
		LOS wind velocity search range	v_{search}	-30 to 30	m/s
		LOS measurement rate	T	0.5*3	s

*1 Requirement in MOPS of DO-220 (The beam scanning speed coincides with the requirements of scanning width (+/- 50 deg) and the required scan time (20 s for a reciprocating scan). The values for the number of range gates (*Ne* and *Na*) are related with the requirement regarding the length of the attended volume (1,000m in MOPS) and the range gate width.
*2 Parameters used for example calculation in Appendix B of DO-220.
*3 Corresponding accumulation number: 500

in Table II. There are many equations that are exactly the same as DO-220, but some changes are due to the difference in characteristics between a CDL and a Doppler radar. The fidelity of the analogy depends on whether the equations are derived only from the general theory of statistics in which the above-mentioned difference does not exist.

IV. SIMULATION RESULTS

A simulation is performed using the parameters of Table I. The relation between SNR, precision of good estimates, and signal detection probability can be known from Fig. 6, by considering the difference in accumulation numbers (100 for Fig. 6, 500 for Table I). The simulation parameters used to generate this figure is the same as shown in Table I. The values of the parameters a'–e' are denoted in the figure. The range dependence on SNR is shown in Fig. 7. The threshold corresponding to the signal detection probability of 80% is also shown in the figure. The range dependences on standard deviation of good estimates and detection probability are shown in Fig. 8. The range dependence on estimation precisions of F-factor and the weighted average of F-factors are shown in Fig. 9. The range dependences on the probabilities regarding the missed and the false alerts for the single LOS are shown in Fig. 10. The same range dependences for multiple LOSs are additionally shown in the figure. Fig. 11 shows the same plots for the multiple scans before and after the additional (i.e., $N(R)$ times) scans. The thresholds of 10^{-5} for the must-alert and 4×10^{-4} for the must-not alert F-factors are also plotted in the figure. These are the thresholds which are the requirements in the MOPS of DO-220. The missed and false alert probabilities of hazardous LLWS detections averaged over the population are shown in Fig. 12. The thresholds of 10^{-5} for the must alert and 10^{-4} for the must-not alert LLWSs are also plotted in the figure. Several observations on these results can be made.

1) The SNR decreases for a longer range (see Fig. 7).
2) The decreased SNR degrades the precision of good estimates of wind velocity and signal detection probability (see Fig. 8).
3) F-factor estimation precision becomes worse for longer ranges but is improved by weighted averaging (see Fig. 9).
4) Continuity judgment for multiple LOSs contributes to a lower missed alert probability but increases the false alert probability (see Fig. 10).
5) The same manner for multiple scans and additional multiple scans. The tendency is more distinct at nearer ranges since the number of additional scans is larger for these ranges (see Fig. 11).
6) The stepwise plots appear in the missed-alert probability for the additional multiple scans because the number of scans changes by each reciprocating motion (therefore, does not change smoothly regarding range) (see Figs. 11 and 12).

Fig. 7 shows that practical CDL parameters in Table I can realize wind sensing for ranges of more than 12 km. Figs. 8–12

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TABLE II
OUTPUT PARAMETERS OF SIMULATION AND CORRESPONDENCE OF EQUATIONS WITH DO-220

Category		Contents of equation	Symbol	Correspondent equation		
				This paper (CDL)	DO-220 [1] (Radar)	Perfectly same?
SNR		Range dependence of SNR	$SNR(R)$	(2) ~ (4)	(1)	No
Wind sensing performance		Range dependence of standard deviation of good estimates and detection probability	$P_{rob}(R)$ $P_{rec}(R)$	(5) ~ (10)	(2)	No
LLWS detection	F-factor precision for single LOS	Range dependence of estimation precision of F-factor	$\sigma_F(SNR(R))$	(11)	(3)	Yes
		Range dependence of estimation precision on weighted averaged F-factor	$\sigma_{\bar{F}}(SNR(R))$	(12)	(4)	Yes
		Range dependence of histogram on weighted averaged F-factor	$PDF_F(F_e, F_m, SNR(R))$	(13)	-	No
	LLWS detection for single LOS	Range dependence of detection probability for must alert F-factor	$PD^S(F_a, SNR(R))$	(14)	(6)	Yes
		Range dependence of missed alert probability for must alert F-factor (for single LOS)	$PM^S(F_a, SNR(R))$	(15)	(7)	Yes
		Range dependence of false alert probability for must-not alert F-factor (for single LOS)	$PF^S(F_{na}, SNR(R))$	(16)	(8)	Yes
	LLWS detection for multiple LOSs	Range dependence of missed detection probability for must alert F-factor (after continuity judgement for multiple LOS)	$PM(F_a, SNR(R))$	(17)	(9)	Yes
		Range dependence of false alert probability for must-not alert F-factor (after continuity judgement for multiple LOSs)	$PF(F_{na}, SNR(R))$	(18)	(10)	Yes
		Number of LOS directions used for the continuity judgement (not numbered in [1] but the concept is perfectly same)	M	(19)	-	No
	LLWS detection for multiple scan	Range dependence of detection probability for must alert F-factor (after continuity judgement in multiple scan)	$PD'(F_a, R)$	(20)	(11)	Yes
		Range dependence of missed alert probability for must alert F-factor (after continuity judgement in multiple scan)	$PM'(F_a, R)$	(21)	(12)	Yes
		Measurement range of kth scan in multiple scan (not shown in [1] but the concept is perfectly same)	R_k	(22)	-	Yes
		Range dependence of false alert probability for must-not alert F-factor (after continuity judgement in multiple scan)	$PF'(F_a, R)$	(23)	(13)	Yes
		Total number of scans observed after approaching to a range R from an approaching hazard	$N(R)$	(24)	(16)	Yes
		Range dependence of missed alert probability for hazardous LLWS (after continuity judgement in multiple scan)	$PM''(F_a, R)$	(25)	(14)	Yes
		Range dependence of false alert probability for non-hazardous LLWS (after continuity judgement in multiple scan)	$PF''(F_{na}, R)$	(26)	(15)	Yes
	LLWS model	Equation for appearance probability of F-factor	$f_H(F)$	(27)	(18)	Yes
	LLWS detection for total performance	Equation for missed alert probability of hazardous LLWS	$\langle PM(R) \rangle$	(28)	(22) ~ (23)	No
		Equation for false alert probability of non-hazardous LLWS	$\langle PF(R) \rangle$	(29)	(24)	No

show the following performances. The missed-alert probability is less than 10^{-5} for the must-alert F-factor, and the false-alert probability is less than 4×10^{-4} for the must-not alert F-factor at a range of 7 km. The probabilities, averaged over the population of LLWS, are less than 10^{-5} for the must alert F-factor and the false-alert probability of less than 10^{-4} for the must-not alert F-factor. These results satisfy the requirements of the MOPS in DO-220. The above-mentioned detection range of 7 km is also larger than the requirement in the MOPS (3378 ft $\approx$ 1.9 km). Further, this result is well-aligned with the detectable range of 6.7–9.5 km in prior LLWS detection experiments [13], even though more track records for demonstrations are needed in the future. The LLWS detection parameters in [13] are the must-alert of 0.13, the must-not alert F-factor of 0.085, and the hazardous F-factor alert threshold of 0.096. These values are almost the same as the ones in Table I of this article.

Figs. 7–12 are the simulation results only for one case, and performances of LLWS detection changes depending on, for example, altitudes. However, simulation results for other cases are not shown here since the most important scope of this article is to clarify how to simulate LLWS detection performances. Further, the aerosol backscatter coefficient, which strongly impacts the performance of a CDL, does not

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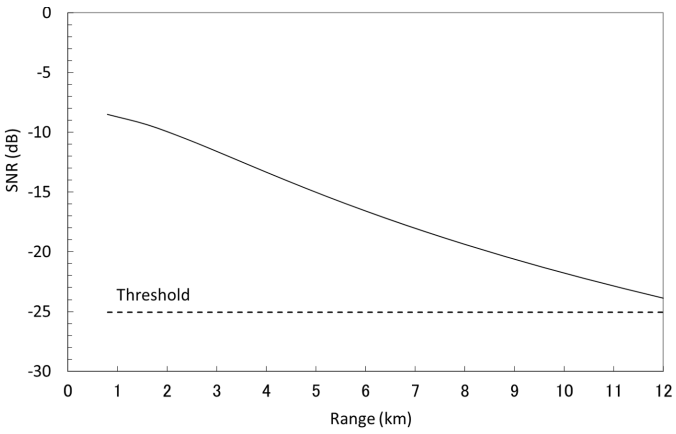


Fig. 7. Range dependence of SNR.

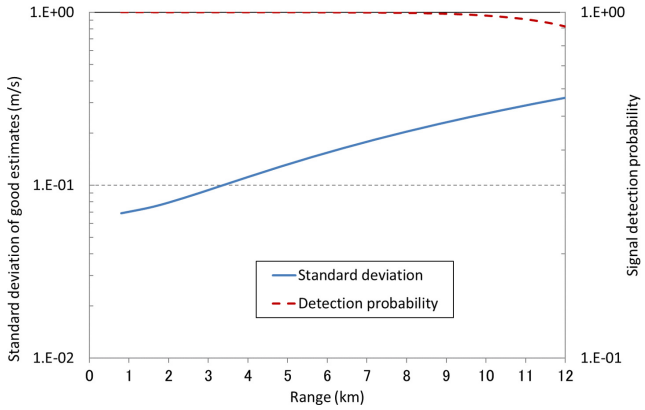


Fig. 8. Range dependences on standard deviation of good estimates and signal detection probability.

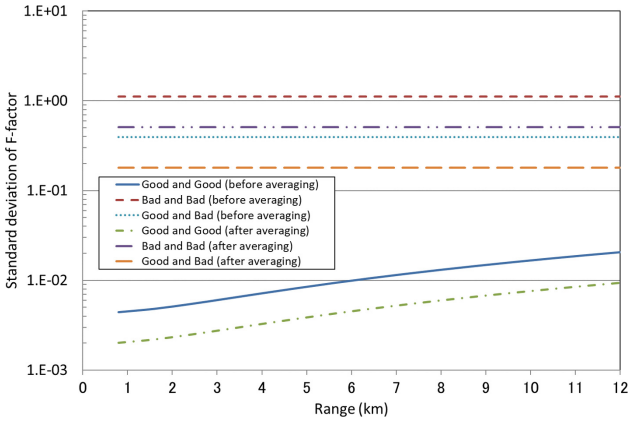


Fig. 9. Range dependences on estimation precisions of F-factor and weighted averaged one for the cases of F-factor calculation with: 1) two good estimates; 2) two bad estimates; and 3) one good estimate and one bad estimate.

change so much within the atmospheric boundary layer in which LLWSs appear. Atmospheric parameters for some other altitudes are shown in Appendix B for reference.

V. Conclusion

The performance simulation theory of LLWS detection using an airborne CDL was shown, using a simulation theory analogous to RTCA DO-220 for airborne Doppler radar [1].

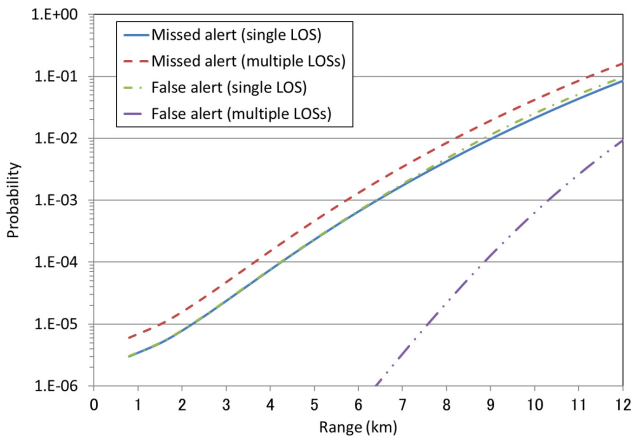


Fig. 10. Range dependences on probabilities regarding missed and false alert for single and multiple LOS.

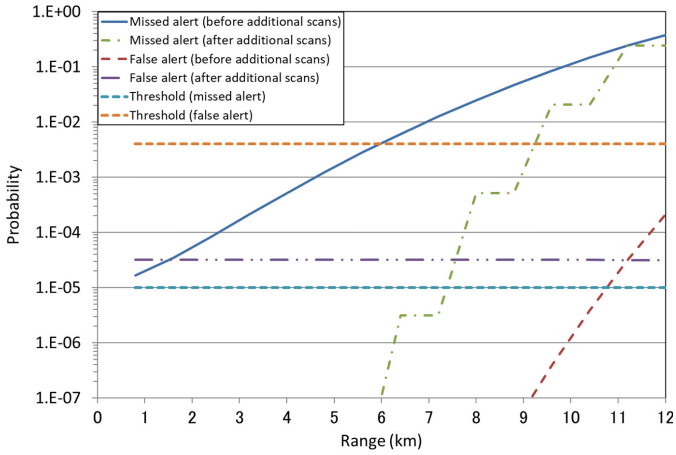


Fig. 11. Same plots as Fig. 10 but for the multiple scans before and after the additional scans.

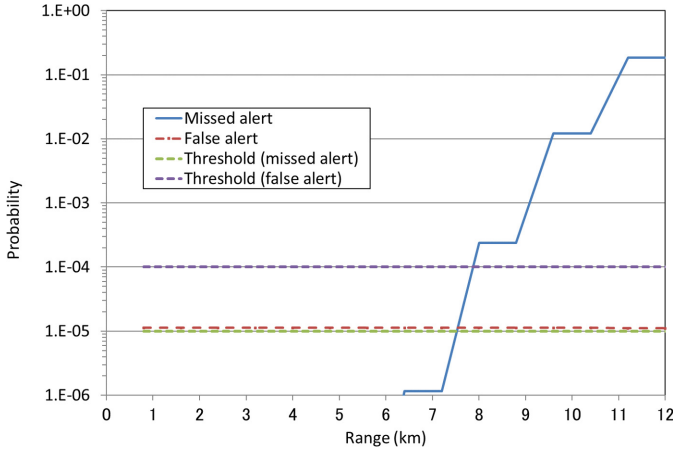


Fig. 12. Missed and false alert probabilities of hazardous LLWS detection averaged over the population.

Prior theoretical studies of CDL on: 1) SNR equation; 2) relation between SNR and wind sensing performance; and 3) atmospheric parameters were fully utilized. After explaining the concept of LLWS detection using an airborne

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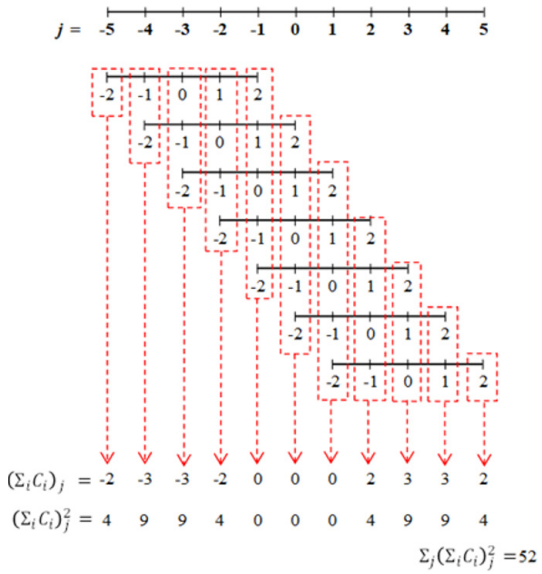


Fig. 13. Schematic of the weighting coefficients in moving average of F-factor.

CDL, the simulation theory on LLWS detection performance was described, and the analogy to the equations in DO-220 was summarized in addition to showing the equations and input–output parameters of the simulation. The simulation procedure coincides with the one in DO-220 but there are some differences in the equations for SNR calculation and wind sensing performance (standard deviation of good estimates and signal detection probability). One set of performance simulation results is introduced based on the actually applied CDL specification that was developed in the context of the SafeAvio project of JAXA [11], [12], [13]. It is shown that wind sensing can be realized for a range of more than 12 km using practical CDL parameters. Furthermore, it realizes a missed-alert probability of less than 10^{-5} for the must-alert F-factor and a false-alert probability of less than 4×10^{-4} for the must-not alert F-factor simultaneously at a range of 7 km. The probabilities averaged over the population of LLWS equally yield a missed-alert probability of less than 10^{-5} for the must-alert F-factor and a false-alert probability of less than 10^{-4} for the must-not alert F-factor. These results satisfy the requirements of the MOPS in DO-220. The detection range also satisfies the requirement in the MOPS. Further, this result agreed with the detectable range of 6.7–9.5 km in prior LLWS detection experiments with the same detection parameters, even though more track records for the demonstration experiments are needed in the future. The results of this study can be utilized to establish MOPS for LLWS detection using an airborne CDLs in the future.

APPENDIX A

The weighted moving average of the F-factor is shown in Fig. 13. The example in the figure shows the case of $N_e = 5$ and $N_a = 7$. The number of range gates which is used to obtain the moving average is $N_e + N_a = 11$. This is equivalent to the case of an averaging width of 1 km with a range gate width of 100 m. The weighting (C_i) is the gate number distance from the center of each set of N_e.

TABLE III
ATMOSPHERIC PARAMETERS FOR EACH ALTITUDE

Altitude (km)	Aerosol backscatter coefficient (/m/sr)	Extinction to aerosol backscatter ratio (sr)	Molecular absorption coefficient (/m)	Atmospheric refractive index structure constant ($m^{-2/3}$)
0.5	1.1×10^{-7}	50	5.4×10^{-6}	8.9×10^{-16}
1	4.0×10^{-8}	50	3.7×10^{-6}	1.5×10^{-16}
3	1.3×10^{-9}	50	1.5×10^{-6}	3.9×10^{-17}
5	5.4×10^{-10}	50	5.1×10^{-7}	1.4×10^{-17}
7	3.4×10^{-10}	50	2.4×10^{-7}	1.6×10^{-17}
10	3.3×10^{-10}	50	8.8×10^{-8}	2.4×10^{-17}
12	2.8×10^{-10}	50	3.1×10^{-8}	2.0×10^{-17}

APPENDIX B

Atmospheric parameters for some altitudes are shown in Table III. The values are for the wavelength of 1.55 μm. The aerosol backscatter coefficient, the lidar ratio, and the atmospheric refractive index structure constant are the same as the ones in [14, Table 5-1]. The values of these parameters have been set under the discussion between JAXA, NASA, Deutsches Zentrum für Luft-und Raumfahrt (DLR), and so on although the sources of these values have not been seen in [14]. The value of aerosol backscatter coefficient at each altitude in [14] corresponds to a cumulative appearance probability of 90%. This means aerosol backscatter coefficients are larger than this value in 90% cases. There are several other literature works which have reported about aerosol backscatter coefficients at upper altitudes (for example, [30], [31], [32], [33], [34], [35]). Therefore, more careful discussion regarding aerosol backscatter coefficients might be needed in future feasibility studies of airborne CDLs for cruising altitude applications (for example, long-range CAT detections). The molecular absorption coefficients have been derived using the U.S. standard atmosphere model.

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